10

a $m\angle UPT = 30^\circ$

b $m\overset{\frown}{US} = 60^\circ$

11 $x = 3$

12

a $x = 20$

b $m\overset{\frown}{SV} = 110^\circ$

13 $m\angle F = 180^\circ - a^\circ$

14

a $m\angle 1 = 60^\circ$

b $m\overset{\frown}{BD} = 168^\circ$

15

a $x = \frac{40}{9}$

b $m\overset{\frown}{GF} = 112^\circ$

16 The angles are congruent because they are all inscribed angles with the same intercepted arc

17 $\angle CAD$ and $\angle CBD$ have the same intercepted arc, $\overset{\frown}{CD}$. Since $\angle CAD$ is a central angle, $m\overset{\frown}{CD} = m\angle CAD$. Since $\angle CBD$ is an inscribed angle, $m\overset{\frown}{CD} = 2(m\angle CBD)$. Using substitution we get $m\angle CAD = 2(m\angle CBD)$, therefore $y = 2x$.

18

a 240°

b $\overset{\frown}{FEA}$, $\overset{\frown}{CDB}$, and $\overset{\frown}{BFE}$

c $m\overset{\frown}{COF} = \frac{1}{2}m\overset{\frown}{CAF}$

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19

a 45°

b 90°

c $\frac{1}{2}\left(\frac{32-2n}{32} * 360\right)^\circ$

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